Sequences & Series

Section 1: Arithmetic Sequences & nth Term

1 In an arithmetic sequence where $a_1 = 5$ and $d = 3$, what is the 12th term a_{12}?

a. 38

b. 41

c. 35

d. 44

2 If the 4th term of an arithmetic sequence is 17 and the 8th term is 33, what is the common difference d?

a. 4

b. 5

c. 3

d. 6

3 What is the sum of the first 10 terms of the arithmetic series: $2 + 5 + 8 + \ldots$?

a. 155

b. 150

c. 160

d. 145

Section 2: Geometric Sequences & Finite Series

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4 What is the 6th term of the geometric sequence $2, 6, 18, 54, \ldots$?

a. 486

b. 162

c. 1458

d. 243

5 In a geometric sequence, $a_1 = 4$ and $r = 2$. What is the sum of the first 5 terms (S_5)?

a. 124

b. 60

c. 120

d. 128

6 What is the common ratio r of a geometric sequence where $a_2 = 6$ and $a_5 = 162$?

a. 3

b. 2

c. 4

d. 9

Section 3: Infinite Geometric Series & Convergence

7 What is the sum of the infinite geometric series: $12 + 6 + 3 + 1.5 + \ldots$?

a. 24

b. 18

c. 20

d. 36

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8 For what values of x does the infinite geometric series $\sum_{k=1}^{\infty} (2x)^k$ converge?

a. $|x| < \frac{1}{2}$

b. $|x| < 1$

c. $|x| < 2$

d. $x > 0$

9 Express the repeating decimal $0.\overline{6} = 0.666\ldots$ as a simplified fraction.

a. $\frac{2}{3}$

b. $\frac{3}{5}$

c. $\frac{6}{11}$

d. $\frac{1}{3}$

Section 4: Sigma Notation & Binomial Theorem

10 Evaluate the summation: $\sum_{k=1}^{5} (3k - 2)$.

a. 35

b. 30

c. 40

d. 25

11 In the expansion of $(x + y)^5$, what is the coefficient of x^3y^2?

a. 10

b. 5

c. 20

d. 15

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12 How many terms are there in the binomial expansion of $(2a - b)^7$?

a. 8

b. 7

c. 9

d. 14

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